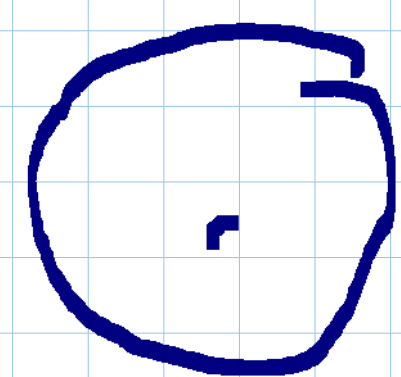


3 Bayntlu 2 paraauk sch Dink m₁ ve m₂ n ver 2 var
 $\Rightarrow \left\{ -\frac{\hbar^2}{2m_1} \nabla_{r_1}^2 - \frac{\hbar^2}{2m_2} \nabla_{r_2}^2 + V(r_1 - r_2) \right\} \Psi = E \Psi$ $r_1 - r_2 = r$ $R = \frac{m_1 + m_2 r_2}{m_1 + m_2}$



$$\Rightarrow \left\{ -\frac{\hbar^2}{2\mu} \nabla_R^2 - \frac{\hbar^2}{2M} \nabla_r^2 + V(r) \right\} \Psi(R, r) = E \Psi(R, r) \Rightarrow \Psi(R, r) = \Psi(R) \cdot R(r) \text{ id.}$$

$$\Rightarrow \text{il } \frac{\hbar^2}{2\mu} \nabla_R^2 \Psi(R) = E_R \Psi(R) \text{ il } \left[\frac{\hbar^2}{2M} \nabla_r^2 + U(r) \right] R(r) = E R(r) \text{ dir}$$

QMI den;

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) R(r) - \frac{2\mu}{\hbar^2} \left[V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] R(r) + \frac{2\mu E}{\hbar^2} R(r) = 0 \text{ id. } V(r) = -\frac{Ze^2}{r} \text{ id.}$$

$$\star \Rightarrow \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) R(r) - \frac{2\mu}{\hbar^2} \left[-\frac{Ze^2}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} + |E| \right] R(r) = 0 \text{ dir.}$$

$$p = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} r \Rightarrow \frac{1}{r} = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{1}{p} \Rightarrow \frac{d}{dr} = \frac{dp}{dr} \cdot \frac{d}{dp} \Rightarrow \frac{d}{dr} = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{d}{dp} \text{ dir.}$$

$$\frac{d^2}{dr^2} = \frac{8\mu E}{\hbar^2} \frac{d^2}{dp^2} \text{ dir}$$

$$\Rightarrow \left(\frac{8\mu E}{\hbar^2} \frac{d^2}{dp^2} + \frac{2}{p} \left(\frac{8\mu E}{\hbar^2} \right) \frac{d}{dp} \right) R(p) - \frac{2\mu}{\hbar^2} \left[-Ze^2 \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{1}{p} + \frac{l(l+1)\hbar^2}{2\mu} \frac{4}{\hbar^2 p^2} + |E| \right] R(p) = 0$$

vt ardi
 $8\mu E / \hbar^2$

paranteze
 alalin

$$\frac{8\mu E}{\hbar^2} \frac{d^2}{dp^2} + \frac{2}{p} \frac{8\mu E}{\hbar^2} \frac{d}{dp} + \left(\frac{2\mu Ze^2}{\hbar^2} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{1}{p} - \frac{2\mu l(l+1)}{\hbar^2} \frac{4}{p^2} - \frac{2\mu |E|}{\hbar^2} \right) R(p) = 0$$

$$\frac{8\mu E}{\hbar^2} \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \frac{8\mu E}{\hbar^2} \left(\frac{2\mu Ze^2}{\hbar^2} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} \frac{1}{p} - \frac{l(l+1)}{p^2 |E|} - \frac{1}{4} \right) R(p) = 0$$

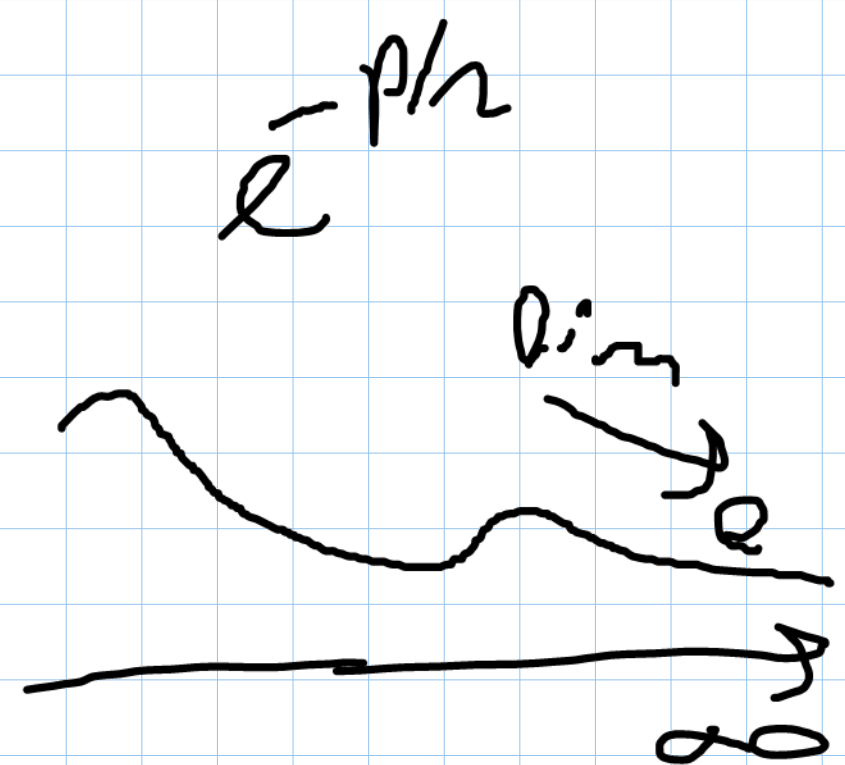
$$\Rightarrow \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \left\{ \left(\frac{p}{2E} \right)^{1/2} \frac{Ze^2}{p\hbar} - \frac{l(l+1)}{p^2 |E|} - \frac{1}{4} \right\} R(p) = 0$$

$$\Rightarrow p = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} r$$

Asynt Davrank $p \rightarrow \infty$ $r \rightarrow \infty$ a glider

$$\Rightarrow \frac{d^2}{dp^2} R(p) - \frac{1}{4} R(p) \equiv 0 \text{ dir } \Rightarrow D^2 - \frac{1}{4} = 0 \Rightarrow D_{1/2} = \pm \frac{1}{2}$$

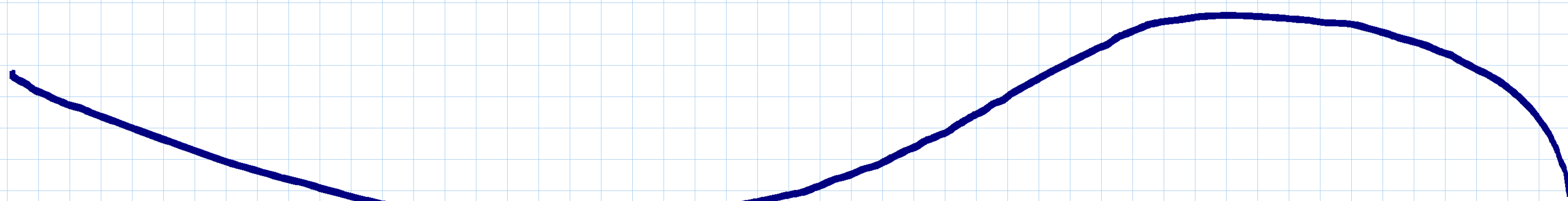
$$\Rightarrow e^{\pm p/2} = R(p) \Rightarrow R'(p) = \pm \frac{1}{2} e^{\pm p/2} \quad R''(p) = +\frac{1}{4} e^{\pm p/2} \quad \checkmark \quad \frac{e^{-p/2}}{e} \lim_{r \rightarrow \infty} = 0$$



$$\Rightarrow \underline{R(p) = G(p) e^{-p/2} \text{ alut.}}$$

$$\Rightarrow R(p) = G(p) e^{-p/2} \Rightarrow R'(p) = G'(p) e^{-p/2} - \frac{1}{2} G(p) e^{-p/2}$$

$$R''(p) = G''(p) e^{-p/2} - \frac{1}{2} G'(p) e^{-p/2} - G'(p) e^{-p/2} + \frac{1}{4} G(p) e^{-p/2}$$



$$\left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp}\right) R(p) + \left\{ \left(\frac{\mu}{2E}\right)^{1/2} \frac{ze^2}{p+h} - \frac{l(l+1)}{p^2|E|} - \frac{1}{4} \right\} R(p)$$

$$R(p) = G(p) e^{-p/2} \Rightarrow R'(p) = G'(p) e^{-p/2} - \frac{1}{2} G(p) e^{-p/2}$$

$$R''(p) = G''(p) e^{-p/2} - \frac{1}{2} G'(p) e^{-p/2} - \frac{1}{2} G'(p) e^{-p/2} + \frac{1}{4} G(p) e^{-p/2}$$

$$\Rightarrow G''(p) e^{-p/2} - G'(p) e^{-p/2} + \frac{1}{4} G(p) e^{-p/2} + \frac{2}{p} G'(p) e^{-p/2} - \frac{2}{p} G(p) e^{-p/2} +$$

$$+ \left\{ \left(\frac{\mu}{2E}\right)^{1/2} \frac{ze^2}{p+h} G(p) e^{-p/2} - \frac{l(l+1)}{p^2|E|} G(p) e^{-p/2} - \frac{1}{4} G(p) e^{-p/2} \right\}$$

$$\Rightarrow \frac{\lambda}{p}$$

$$G''(p) - G'(p) + \frac{1}{4} G(p) + \frac{2}{p} G'(p) - G(p) + \frac{\lambda}{p} G(p) - \frac{l(l+1)}{p^2 E} G(p) - \frac{1}{4} G(p)$$

$$\Rightarrow G'' + G' \left(-1 + \frac{2}{p}\right) + G(p) \left(\frac{1}{4} - 1 + \frac{\lambda}{p} - \frac{l(l+1)}{p^2 E} - \frac{1}{4}\right)$$

$$\Rightarrow G'' + G' \left(-1 + \frac{2}{p}\right) + G \left(\frac{\lambda}{p} - 1 - \frac{l(l+1)}{p^2 E}\right)$$

Probetius jätkami